

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc. I.T Semester V)
Mern Stack
Practical – VIII

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Class: TYIT	Batch: 1
Date of Assignment: 30/09/2026	Date/Time of Submission: 30/09/2026

8. API Creation and Integration

JavaScript normally works on a browser but with node.js we can use it on server.

Express is Library runs on node.js which makes API easy.

Server.js creates express, allow react to communicate, read JSON, connect to MongoDB, create API and start server.

a. Create a simple REST API using Express.

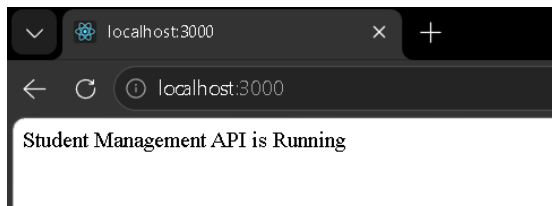
Code:

Server.js:

```
const express = require("express");
const mongoose = require("mongoose");
const cors = require("cors");
const app = express();
const PORT = 3000;
// Middleware
app.use(cors());
app.use(express.json());
// Connect MongoDB
mongoose.connect("mongodb://localhost:27017/CollegeDB") .then(() => {
  console.log("MongoDB Connected Successfully"); })
.catch((error) => {
  console.log("MongoDB Connection Failed");
  console.log(error);
});
// Simple REST API
app.get("/", (req, res) => {
  res.send("Student Management API is Running"); });
app.listen(PORT, () => {
  console.log(`Server is running on http://localhost:${PORT}`); });
```

Output:

```
found 0 vulnerabilities
PS C:\Users\sachin\OneDrive\Dokumen\College all subjects\Practical MERN stack\Practical8> node server.js
Server is running on http://localhost:3000
MongoDB Connected Successfully
```

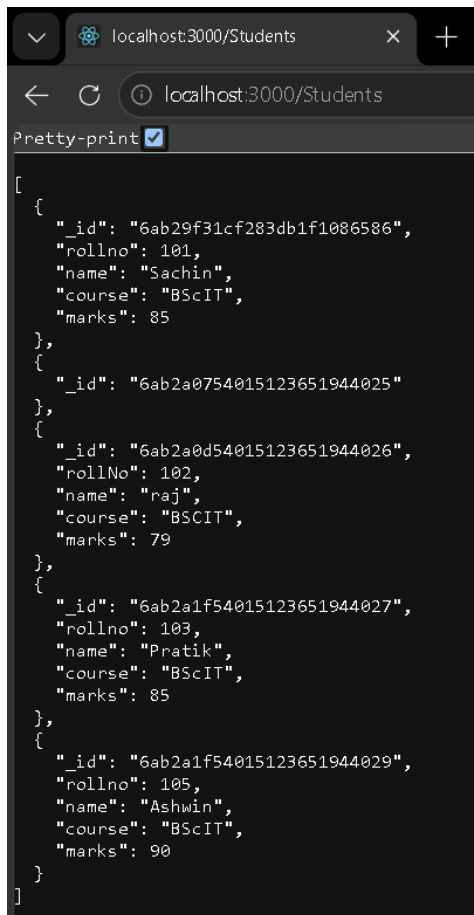


b. Implement a GET API to retrieve data.

Code:

```
const express = require("express");
const mongoose = require("mongoose");
const cors = require("cors");
const app = express();
const PORT = 3000;
// Middleware
app.use(cors());
app.use(express.json());
// Connect MongoDB
mongoose.connect("mongodb://localhost:27017/CollegeDB") .then(() => {
  console.log("MongoDB Connected Successfully"); })
.catch((error) => {
  console.log("MongoDB Connection Failed");
  console.log(error);
});
// Simple REST API
app.get("/", (req, res) => {
  res.send("Student Management API is Running"); });
app.get("/Students", async (req, res) => {
  try {
    const Students = await mongoose.connection.db
      .collection("Students")
      .find()
      .toArray();
    res.json(Students);
  }
  catch (error) {
    res.status(500).json({
      message: "Error fetching students"
    });
  }
});
app.listen(PORT, () => {
  console.log(`Server is running on http://localhost:${PORT}`); });
```

Output:



c. Fetch data from the API in a React application.

Code:

Server.js:

```
const express = require("express");
const mongoose = require("mongoose");
const cors = require("cors");
const app = express();
const PORT = 3000;
// Middleware
app.use(cors());
app.use(express.json());
// Connect MongoDB
mongoose
  .connect("mongodb://127.0.0.1:27017/CollegeDB")
  .then(() => {
    console.log("MongoDB Connected Successfully");
  })
  .catch((error) => {
    console.log("MongoDB Connection Failed");
    console.error(error);
  });

// Home API
app.get("/", (req, res) => {
```

```

    res.send("Student Management API is Running");
  });
// GET API - Fetch students
app.get("/students", async (req, res) => {
  try {
    const students = await mongoose.connection.db
      .collection("Students")
      .find()
      .toArray();

    res.json(students);
  } catch (error) {
    console.error(error);

    res.status(500).json({
      message: "Error fetching students",
      error: error.message
    });
  }
});
// POST API - Store students
app.post("/students", async (req, res) => {
  try {
    const student = req.body;
    await mongoose.connection.db
      .collection("Students")
      .insertOne(student);

    res.json({
      message: "Student added successfully",
      student: student
    });
  } catch (error) {
    console.error(error);

    res.status(500).json({
      message: "Error adding student",
      error: error.message
    });
  }
});
// Start Server
app.listen(PORT, () => {
  console.log(`Server is running on http://localhost:${PORT}`);
});

```

App.jsx:

```

import { useEffect, useState } from "react";
import axios from "axios";
function App() {

```

```

const [students, setStudents] = useState([]);
const [rollNo, setRollNo] = useState("");
const [name, setName] = useState("");
const [course, setCourse] = useState("");
const [marks, setMarks] = useState("");
// Get students
const getStudents = () => {
  axios
    .get("http://localhost:3000/students")
    .then((response) => {
      console.log("Student Data:", response.data);
      setStudents(response.data);
    })
    .catch((error) => {
      console.log("Fetch Error:", error);
    });
};
// Fetch data when page loads
useEffect(() => {
  getStudents();
}, []);
// Add student
const addStudent = () => {
  if (!rollNo || !name || !course || !marks) {
    alert("Please fill all fields");
    return;
  }
  const newStudent = {
    rollNo: Number(rollNo),
    name: name,
    course: course,
    marks: Number(marks)
  };
  axios
    .post("http://localhost:3000/students", newStudent)
    .then((response) => {
      alert(response.data.message);
      // Clear form
      setRollNo("");
      setName("");
      setCourse("");
      setMarks("");
      // Refresh student list
      getStudents();
    })
    .catch((error) => {
      console.log("POST Error:", error);
      alert("Error adding student");
    });
};

```

```

return (
  <div>
    <h1>Student Management System</h1>
    <h2>Add Student</h2>
    <input
      type="number"
      placeholder="Roll No"
      value={rollNo}
      onChange={(e) => setRollNo(e.target.value)}
    />
    <br />
    <br />
    <input
      type="text"
      placeholder="Student Name"
      value={name}
      onChange={(e) => setName(e.target.value)}
    />
    <br />
    <br />
    <input
      type="text"
      placeholder="Course"
      value={course}
      onChange={(e) => setCourse(e.target.value)}
    />
    <br />
    <br />
    <input
      type="number"
      placeholder="Marks"
      value={marks}
      onChange={(e) => setMarks(e.target.value)}
    />
    <br />
    <br />
    <button onClick={addStudent}>
      Add Student
    </button>
    <hr />
    <h2>Student List</h2>
    <table border="1" cellPadding="10">
      <thead>
        <tr>
          <th>Roll No</th>
          <th>Name</th>
          <th>Course</th>
          <th>Marks</th>
        </tr>
      </thead>

```

```

        <tbody>
          {students.map((student) => (
            <tr key={student._id}>
              <td>{student.rollNo}</td>
              <td>{student.name}</td>
              <td>{student.course}</td>
              <td>{student.marks}</td>
            </tr>
          ))}
        </tbody>
      </table>
    </div>
  );
}
export default App;

```

Output:

Roll No	Name	Course	Marks
18	sachin	ittt	1

d. Send data from React to the server using a POST request.

Code:

Server.js:

```

const express = require("express");
const mongoose = require("mongoose");
const cors = require("cors");
const app = express();
const PORT = 3000;
// Middleware
app.use(cors());
app.use(express.json());
// Connect MongoDB
mongoose
  .connect("mongodb://127.0.0.1:27017/CollegeDB")
  .then(() => {
    console.log("MongoDB Connected Successfully");
  })

```

```

    .catch((error) => {
        console.log("MongoDB Connection Failed");
        console.error(error);
    });

// Home API
app.get("/", (req, res) => {
    res.send("Student Management API is Running");
});
// GET API - Fetch students
app.get("/students", async (req, res) => {
    try {
        const students = await mongoose.connection.db
            .collection("Students")
            .find()
            .toArray();

        res.json(students);
    } catch (error) {
        console.error(error);

        res.status(500).json({
            message: "Error fetching students",
            error: error.message
        });
    }
});
// POST API - Store students
app.post("/students", async (req, res) => {
    try {
        const student = req.body;
        await mongoose.connection.db
            .collection("Students")
            .insertOne(student);

        res.json({
            message: "Student added successfully",
            student: student
        });
    } catch (error) {
        console.error(error);

        res.status(500).json({
            message: "Error adding student",
            error: error.message
        });
    }
});
// Start Server
app.listen(PORT, () => {

```



```
    console.log(`Server is running on http://localhost:${PORT}`);  
  });
```

App.jsx:

```
import { useEffect, useState } from "react";  
import axios from "axios";  
function App() {  
  const [students, setStudents] = useState([]);  
  const [rollNo, setRollNo] = useState("");  
  const [name, setName] = useState("");  
  const [course, setCourse] = useState("");  
  const [marks, setMarks] = useState("");  
  // Get students  
  const getStudents = () => {  
    axios  
      .get("http://localhost:3000/students")  
      .then((response) => {  
        console.log("Student Data:", response.data);  
        setStudents(response.data);  
      })  
      .catch((error) => {  
        console.log("Fetch Error:", error);  
      });  
  };  
  // Fetch data when page loads  
  useEffect(() => {  
    getStudents();  
  }, []);  
  // Add student  
  const addStudent = () => {  
    if (!rollNo || !name || !course || !marks) {  
      alert("Please fill all fields");  
      return;  
    }  
    const newStudent = {  
      rollNo: Number(rollNo),  
      name: name,  
      course: course,  
      marks: Number(marks)  
    };  
    axios  
      .post("http://localhost:3000/students", newStudent)  
      .then((response) => {  
        alert(response.data.message);  
        // Clear form  
        setRollNo("");  
        setName("");  
        setCourse("");  
        setMarks("");  
        // Refresh student list
```

```

        getStudents();
    })
    .catch((error) => {
        console.log("POST Error:", error);
        alert("Error adding student");
    });
};
return (
    <div>
        <h1>Student Management System</h1>
        <h2>Add Student</h2>
        <input
            type="number"
            placeholder="Roll No"
            value={rollNo}
            onChange={(e) => setRollNo(e.target.value)}
        />
        <br />
        <br />
        <input
            type="text"
            placeholder="Student Name"
            value={name}
            onChange={(e) => setName(e.target.value)}
        />
        <br />
        <br />
        <input
            type="text"
            placeholder="Course"
            value={course}
            onChange={(e) => setCourse(e.target.value)}
        />
        <br />
        <br />
        <input
            type="number"
            placeholder="Marks"
            value={marks}
            onChange={(e) => setMarks(e.target.value)}
        />
        <br />
        <br />
        <button onClick={addStudent}>
            Add Student
        </button>
        <hr />
        <h2>Student List</h2>
        <table border="1" cellPadding="10">
            <thead>

```

```

        <tr>
          <th>Roll No</th>
          <th>Name</th>
          <th>Course</th>
          <th>Marks</th>
        </tr>
      </thead>
      <tbody>
        {students.map((student) => (
          <tr key={student._id}>
            <td>{student.rollNo}</td>
            <td>{student.name}</td>
            <td>{student.course}</td>
            <td>{student.marks}</td>
          </tr>
        ))}
      </tbody>
    </table>
  </div>
);
}
export default App;

```

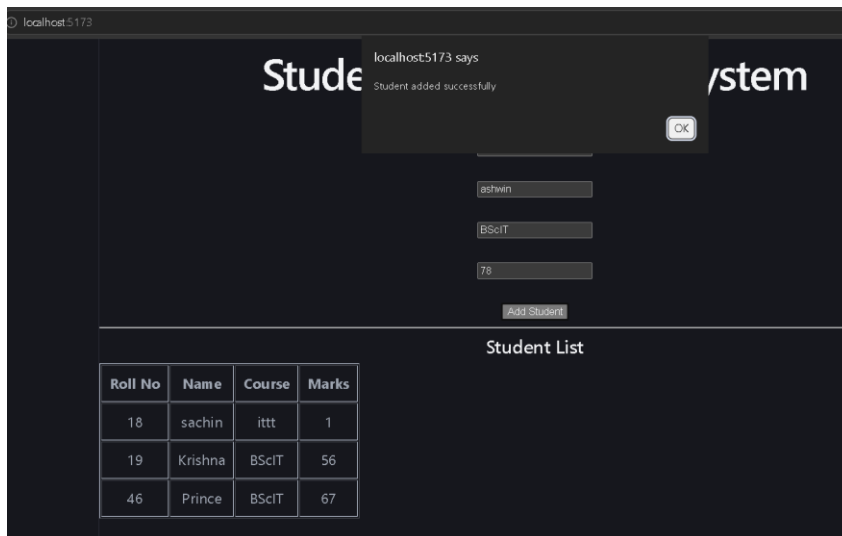
Output:

Student Management System

Add Student

Student List

Roll No	Name	Course	Marks
18	sachin	ittt	1
19	Krishna	BScIT	56
46	Prince	BScIT	67



e. Store data in MongoDB.

Code:

App.jsx:

```
import { useEffect, useState } from "react";
import axios from "axios";
```

```
function App() {

  const [students, setStudents] = useState([]);
  const [rollNo, setRollNo] = useState("");
  const [name, setName] = useState("");
  const [course, setCourse] = useState("");
  const [marks, setMarks] = useState("");

  // Get students
  const getStudents = () => {
    axios
      .get("http://localhost:3000/students")
      .then((response) => {
        console.log("Student Data:", response.data);
        setStudents(response.data);
      })
      .catch((error) => {
        console.log("Fetch Error:", error);
      });
  };

  // Fetch data when page loads
  useEffect(() => {
    getStudents();
  }, []);

  // Add student
  const addStudent = () => {

    if (!rollNo || !name || !course || !marks) {
```

```

        alert("Please fill all fields");
        return;
    }

    const newStudent = {
        rollNo: Number(rollNo),
        name: name,
        course: course,
        marks: Number(marks)
    };

    axios
        .post("http://localhost:3000/students", newStudent)
        .then((response) => {

            alert(response.data.message);

            // Clear form
            setRollNo("");
            setName("");
            setCourse("");
            setMarks("");

            // Refresh student list
            getStudents();
        })
        .catch((error) => {
            console.log("POST Error:", error);
            alert("Error adding student");
        });
    });

    return (
        <div>

            <h1>Student Management System</h1>

            <h2>Add Student</h2>

            <input
                type="number"
                placeholder="Roll No"
                value={rollNo}
                onChange={(e) => setRollNo(e.target.value)}
            />

            <br />
            <br />

            <input

```

```

        type="text"
        placeholder="Student Name"
        value={name}
        onChange={(e) => setName(e.target.value)}
    />

    <br />
    <br />

    <input
        type="text"
        placeholder="Course"
        value={course}
        onChange={(e) => setCourse(e.target.value)}
    />

    <br />
    <br />

    <input
        type="number"
        placeholder="Marks"
        value={marks}
        onChange={(e) => setMarks(e.target.value)}
    />

    <br />
    <br />

    <button onClick={addStudent}>
        Add Student
    </button>

    <hr />

    <h2>Student List</h2>

    <table border="1" cellPadding="10">

        <thead>
            <tr>
                <th>Roll No</th>
                <th>Name</th>
                <th>Course</th>
                <th>Marks</th>
            </tr>
        </thead>

        <tbody>
            {students.map((student) => (

```

```

        <tr key={student._id}>
          <td>{student.rollNo}</td>
          <td>{student.name}</td>
          <td>{student.course}</td>
          <td>{student.marks}</td>
        </tr>
      )}
    </tbody>

  </table>

</div>
);
}

```

export default App;

Server.js:

```

const express = require("express");
const mongoose = require("mongoose");
const cors = require("cors");

const app = express();
const PORT = 3000;

// Middleware
app.use(cors());
app.use(express.json());

// Connect MongoDB
mongoose
  .connect("mongodb://127.0.0.1:27017/CollegeDB")
  .then(() => {
    console.log("MongoDB Connected Successfully");
  })
  .catch((error) => {
    console.log("MongoDB Connection Failed");
    console.error(error);
  });

// Home API
app.get("/", (req, res) => {
  res.send("Student Management API is Running");
});

// GET API - Fetch students
app.get("/students", async (req, res) => {
  try {
    const students = await mongoose.connection.db
      .collection("Students")

```

```

        .find()
        .toArray();

    res.json(students);
  } catch (error) {
    console.error(error);

    res.status(500).json({
      message: "Error fetching students",
      error: error.message
    });
  }
});

// POST API - Store students
app.post("/students", async (req, res) => {
  try {
    const student = req.body;

    await mongoose.connection.db
      .collection("Students")
      .insertOne(student);

    res.json({
      message: "Student added successfully",
      student: student
    });
  } catch (error) {
    console.error(error);

    res.status(500).json({
      message: "Error adding student",
      error: error.message
    });
  }
});

// Start Server
app.listen(PORT, () => {
  console.log(`Server is running on http://localhost:${PORT}`);
});

```

Output:

CONNECTIONS (1)

×

+

...

Search connections

▼

▼

localhost:27017

▼

CollegeDB

▼

Students

▼

students

▼

admin

▼

college_canteen

▼

config

▼

local

🕒 ▼

Type a query: { field: 'value' } or [Generate query](#) 🚀

➕ ADD DATA ▼

📄 BULK ▼

📄 EXPORT DATA ▼

📄 EXPORT CODE

```
_id: ObjectId('6abcb0292cab3c9841e0348f')
rollNo: 18
name: "sachin"
course: "ittt"
marks: 1
```

f. Display stored data in React page.

Student Management System

Add Student

Roll No

Student Name

Course

Marks

Add Student

Student List

Roll No	Name	Course	Marks
18	sachin	ittt	1